Soal Latihan UTS Mat Dis: Variant

Same questions as `Soal latihan UTS Mat Dis 20260925.docx`, with new numbers. All the data in this version is consistent. Answers are in the separate file *UTS Practice Quiz Variant - Answer Key.pdf*.

Question 1: Boolean Expression, Karnaugh Map, and Logic Circuit

Consider a digital system with three input variables P, Q, and R, and one output S. The behavior of the system is given by the following truth table:

P	Q	R	S
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	1

Based on the truth table:

a. (5%) Write the Boolean function S(P,Q,R) in canonical Sum-of-Products (SOP) form.
b. (5%) Express the function using minterm notation.
c. (5%) Construct the 3-variable Karnaugh map for the function.
d. (5%) Use the Karnaugh map to obtain the simplest Boolean expression.
e. (10%) Draw the digital logic circuit corresponding to the simplified Boolean expression.

Question 2: Set Theory

A lecturer surveys 120 students to determine which subjects they know. Let:

- M = students who know Math
- S = students who know Science
- L = students who know Literature

The survey gives the following information:

$|M| = 44$, $|S| = 28$, $|L| = 31$
$|M \cap L| = 14$, $|M \cap S| = 16$, $|S \cap L| = 11$
and 68 students know at least one of the three subjects.

Answer the following:

a. (5%) How many students know all three subjects?
b. (5%) How many students know none of the three subjects?
c. (5%) How many students know Math and Science but not Literature?
d. (5%) How many students know only Science?

Question 3: Fuzzy Set

Let the universe of discourse be U = {10, 20, 30, 40, 50}. Two fuzzy sets A and B are defined by:

A = {0.3/10, 0.9/20, 0/30, 0.6/40, 0.4/50}
B = {0.5/10, 0.2/20, 0.7/30, 0.6/40, 1/50}

Determine:

a. (5%) $A \cup B$
b. (5%) $A \cap B$
c. (5%) The complement A^c
d. (5%) The support of A
e. (5%) The 0.4-cut of A, denoted $A_{0.4}$

Question 4: Properties of Relations

Let A = {3, 6, 7, 12, 18} and define a relation R on A by

$aRb \iff a \mid b$

where $a \mid b$ means "a divides b."

Answer the following questions:

a. (5%) Write R explicitly as a set of ordered pairs.
b. (5%) Determine whether R is reflexive, symmetric, antisymmetric, and transitive. Give a brief justification for each answer.
c. (5%) Based on your answers in part (b), determine whether R is a partial order relation.
d. (5%) Draw the Hasse diagram of (A, R).
e. (5%) Identify the minimal element(s), maximal element(s), least element, and greatest element.